MODULE 3 NOTES

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1. FOUNDATIONS OF MACHINE LEARNING

1.1 Need for Machine Learning

Traditional computer programs are developed by explicitly specifying rules and instructions.

Traditional Programming

Input Data + Rules → Output

For example, to identify whether an email is spam, a programmer might create rules such as:

- If the email contains "free money", mark it as spam.
- If the email contains many suspicious links, mark it as spam.
- If the sender is unknown, increase the spam score.

However, creating rules manually becomes difficult when the problem is complex.

Machine Learning Approach

Machine learning allows the computer to learn patterns automatically from examples.

Training Data → Learning Algorithm → Model

The model can then be used on new data.

Why is Machine Learning Needed?

1. **Handling complex problems**
    - Some relationships are difficult to express using explicit rules.
    - Example: recognizing faces in images.
2. **Large amounts of data**
    - Organizations generate huge quantities of data.
    - ML can identify useful patterns in large datasets.

3. **Automation**
    - ML can automate repetitive prediction and classification tasks.
4. **Prediction**
    - Historical data can be used to predict future outcomes.
    - Example: predicting house prices.
5. **Personalization**
    - Recommendation systems can learn individual user preferences.
6. **Adaptability**
    - ML models can be retrained when new data becomes available.
7. **Pattern discovery**
    - ML can identify hidden relationships in data.

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2. MACHINE LEARNING EXPLAINED

Definition

Machine Learning is a branch of Artificial Intelligence that enables computers to learn patterns from data and improve their performance on a task without being explicitly programmed for every individual situation.

A machine-learning system generally consists of:

Data → Features → Learning Algorithm → Model → Prediction/Decision

Example: House Price Prediction

Suppose we have:

Area	Bedrooms	Age	Price
1200	2	10	₹50 lakh
1500	3	5	₹70 lakh
2000	4	3	₹95 lakh

The first three columns are **features**, while Price is the **target variable**.

The ML algorithm learns a relationship between the features and price.

For a new house:

Area = 1800 sq. ft., Bedrooms = 3, Age = 4

the trained model predicts the price.

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3. MACHINE LEARNING AND OTHER FIELDS

Machine Learning is closely related to several disciplines.

Field	Relationship with Machine Learning
Artificial Intelligence	ML is one of the major approaches used to build intelligent systems
Statistics	Provides probability, estimation, inference and statistical modeling
Data Science	Uses ML along with statistics, programming, visualization and domain knowledge
Linear Algebra	Provides vectors, matrices, eigenvalues and matrix operations
Calculus	Used in optimization and gradient-based learning
Optimization	Helps find model parameters that minimize error
Database Systems	Provide storage and retrieval of training data
Pattern Recognition	Focuses on identifying patterns in data

AI vs ML vs Data Science

Artificial Intelligence

→ Broad field concerned with intelligent behavior.

Machine Learning

→ A major approach for learning patterns from data.

Data Science

→ Broader data-oriented discipline involving:

Data Collection + Statistics + Programming + Visualization + ML + Domain Knowledge

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4. TYPES OF MACHINE LEARNING

The major types are:

1. Supervised Learning
2. Unsupervised Learning
3. Semi-Supervised Learning
4. Reinforcement Learning

4.1 Supervised Learning

In supervised learning, the algorithm learns from **labeled data**.

Structure

Input X → Learning Algorithm → Target Y

The training dataset contains both input and expected output.

Major tasks

Classification

Output is a category.

Examples:

- Spam / Not Spam
- Disease / No Disease
- Cat / Dog
- Pass / Fail

Regression

Output is a numerical value.

Examples:

- House price
- Temperature
- Salary
- Sales prediction

Examples of supervised algorithms

- Linear Regression

- Logistic Regression
- Decision Tree
- Random Forest
- K-Nearest Neighbors
- Support Vector Machine
- Neural Networks

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5. UNSUPERVISED LEARNING

In unsupervised learning, the data does not contain predefined target labels.

The algorithm attempts to discover hidden structures.

Major tasks

Clustering

Groups have similar observations.

Example:

Customer segmentation:

- Group 1 → High spending
- Group 2 → Medium spending
- Group 3 → Low spending

Dimensionality Reduction

Reduces the number of variables.

Examples:

- PCA
- SVD

Applications

- Customer segmentation
- Anomaly detection
- Data visualization
- Feature representation
- Pattern discovery

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6. SEMI-SUPERVISED LEARNING

Semi-supervised learning uses:

Small amount of labeled data + Large amount of unlabeled data

This is useful when obtaining labels is expensive.

Example

Suppose an organization has:

- 1,000 labeled medical images
- 100,000 unlabeled medical images

Semi-supervised learning can use both datasets.

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7. REINFORCEMENT LEARNING

Reinforcement learning involves an **agent** interacting with an environment.

The agent performs actions and receives rewards or penalties.

Basic structure

Agent → Action → Environment → Reward → Agent

Examples:

- Robot navigation
- Game playing
- Autonomous systems
- Resource allocation

The objective is generally to learn a policy that maximizes cumulative reward.

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8. CHALLENGES OF MACHINE LEARNING

8.1 Poor Quality Data

Data may contain:

- Missing values
- Incorrect values
- Duplicates
- Noise
- Inconsistent formats

Principle

Garbage In → Garbage Out

Poor-quality training data can produce poor models.

8.2 Overfitting

Overfitting occurs when a model learns the training data too closely, including noise.

Characteristics

- Very high training performance
- Poor test performance

Solutions

- More training data
- Regularization
- Feature selection
- Cross-validation
- Simpler model

8.3 Underfitting

Underfitting occurs when a model is too simple to capture important patterns.

Characteristics

- Poor training performance
- Poor test performance

Solutions

- More appropriate model complexity
- Better features
- Reduce excessive regularization
- Train for longer where appropriate

8.4 High Dimensionality

A dataset may contain hundreds or thousands of features.

Problems include:

- Increased computation
- More storage
- Redundant features
- Noise
- Overfitting
- Curse of dimensionality

Solutions include:

- Feature selection
- PCA
- LDA
- SVD

8.5 Data Leakage

Data leakage occurs when information that should not be available during training is used by the model.

Example:

Using the final examination score as a feature to predict whether a student will pass the examination.

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9. MACHINE LEARNING PROCESS

A typical ML workflow is:

Problem Definition

↓

Data Collection

↓

Data Understanding

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Data Preprocessing

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Exploratory Data Analysis

↓

Feature Engineering

↓

Feature Selection / Dimensionality Reduction

↓

Train/Test Split

↓

Model Training

↓

Model Evaluation

↓

Hyperparameter Tuning

↓

Deployment

↓

Monitoring and Maintenance

Step 1: Problem Definition

Clearly identify:

- What needs to be predicted?
- Who will use the result?
- What is the target variable?
- What performance metric should be used?

Step 2: Data Collection

Data may come from:

- Databases
- Sensors
- Websites
- Surveys
- APIs
- Transactions
- Applications
- Public datasets

Step 3: Data Preprocessing

Includes:

- Missing-value handling
- Duplicate removal
- Encoding
- Scaling
- Outlier analysis

Step 4: Exploratory Data Analysis

EDA helps understand:

- Distributions
- Relationships
- Outliers
- Patterns
- Trends

Step 5: Model Training

The algorithm learns model parameters from training data.

Step 6: Evaluation

The model is tested using appropriate metrics.

Step 7: Deployment

The model is integrated into a real application.

Step 8: Monitoring

Model performance should be monitored because data distributions and real-world conditions can change.

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10. MACHINE LEARNING APPLICATIONS

Healthcare

- Disease prediction
- Medical image analysis
- Drug discovery
- Patient monitoring

Finance

- Fraud detection
- Credit risk
- Financial forecasting

Retail

- Recommendation systems
- Customer segmentation
- Demand forecasting

Manufacturing

- Predictive maintenance
- Defect detection
- Process optimization

Transportation

- Traffic prediction
- Route optimization
- Driver assistance

Cybersecurity

- Intrusion detection
- Malware detection

- Anomaly detection

Natural Language Processing

- Chatbots
- Sentiment analysis
- Translation
- Text classification

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11. UNDERSTANDING DATA

What is Data?

Data consists of observations, measurements, facts, records, or representations collected about objects, people, events, or processes.

Example

Student dataset:

Student	Attendance	Study Hours	Internal Marks	Result
S1	90	5	45	Pass
S2	75	3	32	Pass
S3	50	1	20	Fail

Here:

- Student → Identifier
- Attendance → Feature
- Study Hours → Feature
- Internal Marks → Feature
- Result → Target

---

12. TYPES OF DATA

Numerical Data

Represents quantities.

Examples:

- Age
- Height
- Salary
- Temperature

Discrete

Countable values.

Example:

Number of students = 40.

Continuous

Can take values within a range.

Example:

Height = 172.5 cm.

Categorical Data

Represents groups or categories.

Examples:

- Gender
- Department
- Color
- City

Nominal

No natural ordering.

Example:

Red, Blue, Green.

Ordinal

Categories have an order.

Example:

Low < Medium < High.

---

13. BIG DATA ANALYTICS

Big Data refers to data with characteristics that make conventional processing difficult.

Five Vs of Big Data

1. Volume

Amount of data.

Example:

Terabytes or petabytes of records.

2. Velocity

Speed at which data is generated.

Example:

Real-time sensor data.

3. Variety

Different forms of data:

- Structured
- Semi-structured
- Unstructured

4. Veracity

Quality and reliability of data.

5. Value

Useful information obtained from data.

---

14. TYPES OF BIG DATA ANALYTICS

14.1 Descriptive Analytics

Answers:

"What happened?"

Examples:

- Monthly sales report
- Number of website visitors

14.2 Diagnostic Analytics

Answers:

"Why did it happen?"

Examples:

- Why did sales decrease?
- Why did website traffic increase?

14.3 Predictive Analytics

Answers:

"What is likely to happen?"

Examples:

- Predict next month's sales.
- Predict customer churn.

14.4 Prescriptive Analytics

Answers:

"What action should be considered?"

Examples:

- Which route should be selected?
- How much inventory should be ordered?

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15. BIG DATA ANALYSIS FRAMEWORK

A general framework is:

1. Data Collection

Gather data from different sources.

2. Data Storage

Store data using appropriate systems.

3. Data Preprocessing

Clean and transform data.

4. Data Analysis

Apply statistical and ML techniques.

5. Visualization

Present patterns through graphs and dashboards.

6. Decision Making

Use the findings to support decisions.

7. Feedback

Use new information to improve the process.

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16. DATA COLLECTION AND PREPROCESSING

Data Collection Sources

Primary Data

Collected specifically for the current study.

Examples:

- Surveys
- Experiments
- Interviews
- Sensor measurements

Secondary Data

Previously collected data.

Examples:

- Government datasets
- Company databases
- Research datasets
- Public repositories

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17. DATA PREPROCESSING

Preprocessing converts raw data into a suitable form for machine learning.

Main Steps

1. Data Cleaning

Identify:

- Missing values
- Duplicates
- Incorrect values
- Inconsistent values

2. Data Integration

Combine data from multiple sources.

3. Data Transformation

Convert data into suitable formats.

4. Data Reduction

Reduce unnecessary information.

5. Data Encoding

Convert categorical variables into numerical representations.

---

18. HANDLING MISSING VALUES

Suppose:

Age	Salary
25	30000
30	?
35	50000

The missing salary may be handled using:

Mean Imputation

Replace with average.

Median Imputation

Replace with median.

Median is often useful when data contains extreme values.

Mode Imputation

Used mainly for categorical variables.

Model-Based Imputation

Use other variables to predict the missing value.

---

19. DATA SCALING

Some ML algorithms are sensitive to differences in feature magnitude.

Example:

- Age: 20–60
- Salary: 20,000–200,000

Salary has much larger numerical values.

Standardization

z=x−μσz = \frac{x-\mu}{\sigma}

where:

- x = original value
- μ = mean
- σ = standard deviation

After standardization, the feature generally has approximately:

Mean = 0

Standard deviation = 1

Min-Max Normalization

x′=x−xminxmax−xminx'=\frac{x-x_{min}}{x_{max}-x_{min}}

Usually transforms values into the range:

0 to 1

---

20. DESCRIPTIVE STATISTICS

Descriptive statistics summarize data.

Main measures include:

Measures of Central Tendency

- Mean
- Median
- Mode

Measures of Dispersion

- Range
- Variance
- Standard deviation
- IQR

Measures of Position

- Percentiles
- Quartiles

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21. MEAN

The arithmetic mean is:

x¯=∑i=1nxin\bar{x}=\frac{\sum_{i=1}^{n}x_i}{n}

Example:

Data:

10, 20, 30, 40, 50

Mean:

10+20+30+40+505=30\frac{10+20+30+40+50}{5}=30

Limitation

Mean is sensitive to extreme values.

---

22. MEDIAN

Median is the middle value after sorting the observations.

Example:

10, 20, 30, 40, 50

Median = **30**

For:

10, 20, 30, 40

Median:

20+302=25\frac{20+30}{2}=25

Median is more resistant to extreme values than the mean.

---

23. MODE

Mode is the most frequently occurring value.

Example:

2, 3, 3, 4, 5

Mode = **3**

Mode is particularly useful for categorical data.

---

24. VARIANCE

Variance measures the spread of observations around the mean.

Population variance:

σ2=∑(xi−μ)2N\sigma^2=\frac{\sum(x_i-\mu)^2}{N}

Sample variance:

s2=∑(xi−x¯)2n−1s^2=\frac{\sum(x_i-\bar{x})^2}{n-1}

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25. STANDARD DEVIATION

Standard deviation is the square root of variance.

σ=σ2\sigma=\sqrt{\sigma^2}

A larger standard deviation indicates greater dispersion around the mean.

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26. RANGE

Range=Maximum−MinimumRange=Maximum-Minimum

Example:

Data:

10, 20, 30, 40, 50

Range:

50−10=4050-10=40

---

27. QUARTILES AND IQR

Quartiles divide ordered data into four parts.

- Q1 = 25th percentile
- Q2 = Median
- Q3 = 75th percentile

Interquartile Range

IQR=Q3−Q1IQR=Q3-Q1

Potential Outlier Rule

Lower boundary:

Q1−1.5(IQR)Q1-1.5(IQR)

Upper boundary:

Q3+1.5(IQR)Q3+1.5(IQR)

Values outside these limits may be flagged as potential outliers.

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28. UNIVARIATE DATA ANALYSIS

Univariate analysis considers **one variable at a time**.

Numerical variables

Common techniques:

- Mean
- Median
- Standard deviation
- Histogram
- Box plot
- Density plot

Categorical variables

Common techniques:

- Frequency table
- Percentage
- Bar chart

---

29. DATA VISUALIZATION

Histogram

Used to examine the distribution of numerical data.

Useful for identifying:

- Center
- Spread
- Skewness
- Multiple peaks
- Potential unusual observations

Box Plot

Shows:

- Minimum
- Q1

- Median
- Q3
- Maximum
- Potential outliers

Bar Chart

Used mainly for categorical data.

---

30. BIVARIATE DATA ANALYSIS

Bivariate analysis studies **two variables simultaneously**.

Examples:

- Study hours vs marks
- Experience vs salary
- Temperature vs electricity consumption

Numerical + Numerical

Use:

- Scatter plot
- Covariance
- Correlation
- Regression

Scatter Plot

Shows the relationship between two numerical variables.

Possible patterns:

- Positive relationship
- Negative relationship
- No obvious relationship
- Nonlinear relationship

---

31. COVARIANCE

Covariance indicates how two variables vary together.

Cov(X,Y)=∑(xi−x¯)(yi−y¯)n−1Cov(X,Y)=\frac{\sum(x_i-\bar{x})(y_i-\bar{y})}{n-1}

Interpretation

Positive covariance:

Both variables tend to increase together.

Negative covariance:

One tends to increase when the other decreases.

Near-zero covariance:

Little linear co-movement, although nonlinear relationships may still exist.

---

32. CORRELATION

Pearson correlation is:

r=Cov(X,Y)sXsYr=\frac{Cov(X,Y)}{s_Xs_Y}

Its value lies between:

−1≤r≤1-1 \leq r \leq 1

Interpretation

Correlation	General interpretation
+1	Perfect positive linear association
0	No linear association
−1	Perfect negative linear association

Important: Correlation does not prove causation.

---

33. MULTIVARIATE DATA

Multivariate data contains multiple variables for each observation.

Example:

Student	Attendance	Study Hours	Assignment	Internal Marks	Final Marks
S1	90	5	85	45	90
S2	80	4	75	40	80
S3	60	2	55	30	60

Several variables are considered simultaneously.

---

34. MULTIVARIATE STATISTICS

Multivariate statistical techniques analyze multiple variables together.

Important techniques include:

- Multiple regression
- MANOVA
- Covariance analysis
- Correlation matrices
- PCA
- LDA
- Cluster analysis

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35. COVARIANCE MATRIX

For multiple variables, pairwise covariance values can be represented in a matrix.

For three variables:

S=[Var(X1)Cov(X1,X2)Cov(X1,X3)Cov(X2,X1)Var(X2)Cov(X2,X3)Cov(X3,X1)Cov(X3,X2)Var(X3)]S= \begin{bmatrix} Var(X_1)&Cov(X_1,X_2)&Cov(X_1,X_3)\\ Cov(X_2,X_1)&Var(X_2)&Cov(X_2,X_3)\\ Cov(X_3,X_1)&Cov(X_3,X_2)&Var(X_3) \end{bmatrix}

Important

- Diagonal → Variances
- Off-diagonal → Covariances

The covariance matrix is fundamental to PCA.

---

36. CURSE OF DIMENSIONALITY

The curse of dimensionality refers to difficulties that arise as the number of features increases.

Problems include:

- Data becomes sparse.
- Distance measures become less informative.
- Computation increases.
- More training data may be required.
- Risk of overfitting increases.

Solutions

- Feature selection
- PCA
- LDA
- SVD
- Regularization
- Better data collection

---

37. LINEAR SYSTEMS

A system of linear equations can be represented as:

Ax=bAx=b

where:

- A = coefficient matrix
- x = unknown vector
- b = output vector

Example:

2x+y=52x+y=5 x−y=1x-y=1

Matrix form:

[211−1][xy]=[51]\begin{bmatrix} 2&1\\ 1&-1 \end{bmatrix} \begin{bmatrix} x\\y \end{bmatrix} = \begin{bmatrix} 5\\1 \end{bmatrix}

Linear algebra is fundamental to machine learning because datasets are commonly represented using matrices.

---

38. GAUSSIAN ELIMINATION

Gaussian elimination is a method for solving systems of linear equations.

Basic procedure

1. Write the augmented matrix.
2. Select a pivot.
3. Eliminate values below the pivot.
4. Continue for subsequent columns.
5. Obtain row-echelon form.
6. Use back substitution.

Example

Consider:

2x+y=52x+y=5 x−y=1x-y=1

Augmented matrix:

[21∣51−1∣1]\begin{bmatrix} 2&1&|&5\\ 1&-1&|&1 \end{bmatrix}

Interchange rows:

[1−1∣121∣5]\begin{bmatrix} 1&-1&|&1\\ 2&1&|&5 \end{bmatrix}

Perform:

R2=R2−2R1R_2=R_2-2R_1

giving:

[1−1∣103∣3]\begin{bmatrix} 1&-1&|&1\\ 0&3&|&3 \end{bmatrix}

Therefore:

y=1y=1

and:

x=2x=2

---

39. IMPORTANCE OF LINEAR ALGEBRA IN ML

Linear algebra is used in:

- Linear regression
- Logistic regression
- Neural networks
- PCA
- LDA
- SVD
- Recommendation systems
- Computer vision
- Natural language processing

Important concepts include:

- Vectors
- Matrices
- Matrix multiplication
- Transpose
- Rank
- Eigenvalues
- Eigenvectors
- Orthogonality
- Matrix decomposition

---

40. MATRIX DECOMPOSITIONS

Matrix decomposition represents a matrix as a product of simpler matrices.

Important decompositions:

LU Decomposition

A=LUA=LU

where:

- L = lower triangular matrix
- U = upper triangular matrix

QR Decomposition

A=QRA=QR

where:

- Q = orthogonal matrix
- R = upper triangular matrix

Used in:

- Least-squares problems
- Numerical computation

Eigenvalue Decomposition

For suitable matrices:

A=QΛQ−1A=Q\Lambda Q^{-1}

For symmetric matrices:

A=QΛQTA=Q\Lambda Q^T

It is closely related to PCA.

Singular Value Decomposition

A=UΣVTA=U\Sigma V^T

SVD is especially important for dimensionality reduction.

---

41. MACHINE LEARNING AND PROBABILITY

Machine learning deals with uncertainty.

For example:

A model may predict:

Probability of spam = 0.92

rather than simply saying:

Spam = Yes

Probability provides a mathematical framework for uncertainty.

---

42. BASIC PROBABILITY

Probability of event A:

$0 \leq P(A) \leq 1$

Conditional Probability

$P(A|B) = \frac{P(A \cap B)}{P(B)}$

It represents the probability of A given that B has occurred.

---

43. BAYES' THEOREM

Bayes' theorem:

$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$

where:

- P(A|B) = posterior probability
- P(B|A) = likelihood
- P(A) = prior probability
- P(B) = evidence

Bayesian reasoning is important in:

- Naive Bayes classification
- Medical diagnosis
- Spam detection
- Risk analysis

---

44. RANDOM VARIABLES

A random variable represents numerical outcomes of a random process.

Discrete Random Variable

Takes countable values.

Example:

Number of defective products.

Continuous Random Variable

Takes values over a continuous range.

Example:

Height, weight, temperature.

---

45. PROBABILITY DISTRIBUTIONS

Important distributions include:

Bernoulli Distribution

Models one binary outcome.

Example:

Pass/Fail.

Binomial Distribution

Models the number of successes in repeated Bernoulli trials.

Normal Distribution

Bell-shaped continuous distribution characterized by:

- Mean
- Variance

Many statistical methods assume or approximate normality under suitable conditions.

---

46. IMPORTANCE OF STATISTICS IN MACHINE LEARNING

Statistics helps with:

- Data summarization
- Sampling
- Estimation
- Hypothesis testing
- Feature analysis
- Model evaluation
- Uncertainty estimation
- Experimental design

Machine learning combines:

Data + Statistics + Algorithms + Optimization + Computing

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47. FEATURE ENGINEERING

Feature engineering is the process of transforming raw data into useful features for machine learning.

Example

Original:

Date = 28-09-2026

Possible features:

- Day = 28
- Month = 9
- Year = 2026
- Day of week
- Weekend = Yes/No

---

48. COMMON FEATURE ENGINEERING METHODS

48.1 Mathematical Transformation

Examples:

x2x^2 log(x)\log(x) x\sqrt{x}

Useful when the relationship between variables and target is nonlinear or highly skewed.

48.2 Aggregation

Example:

For a customer:

- Total purchases
- Average purchase amount
- Number of transactions
- Maximum purchase

48.3 Interaction Features

Suppose:

- Study Hours
- Attendance

An interaction feature might represent:

StudyHours×AttendanceStudyHours\times Attendance

Such features can represent combined effects when justified.

---

49. FEATURE SELECTION

Feature selection means choosing a subset of the original features.

Why feature selection?

- Reduce complexity
- Improve computational efficiency
- Remove irrelevant variables
- Reduce redundancy
- Reduce overfitting
- Improve interpretability

Examples:

- Forward Selection
- Backward Elimination

---

50. STEPWISE FORWARD SELECTION

Forward selection starts with **no features**.

Procedure

1. Start with an empty model.
2. Consider each available feature.
3. Add the feature that provides the greatest improvement according to a predefined criterion.
4. Repeat.
5. Stop when adding features no longer provides sufficient improvement.

Example

Suppose features are:

X1, X2, X3, X4, X5

Process:

Start → {}

↓

Add X2

↓

{X2}

↓

Add X4

↓

{X2, X4}

↓

Add X1

↓

{X2, X4, X1}

↓

Stop when the stopping criterion is reached.

---

51. STEPWISE BACKWARD ELIMINATION

Backward elimination starts with **all features**.

Suppose:

{X1, X2, X3, X4, X5}

Procedure

1. Start with all features.
2. Identify the least useful feature according to a selected criterion.
3. Remove it.
4. Refit/evaluate.
5. Continue until the stopping condition is reached.

Example:

{X1,X2,X3,X4,X5}

↓

Remove X5

↓

{X1,X2,X3,X4}

↓

Remove X3

↓

{X1,X2,X4}

↓

Continue until the criterion indicates stopping.

Limitation

Stepwise procedures can be unstable and may select different variables when the data changes slightly. Proper validation is therefore important.

---

52. DIMENSIONALITY REDUCTION

Dimensionality reduction transforms high-dimensional data into a smaller number of dimensions.

Example

Original:

100 features

↓

Dimensionality reduction

↓

10 components

The goal is to retain useful information while reducing complexity.

---

53. FEATURE SELECTION VS DIMENSIONALITY REDUCTION

Feature Selection	Dimensionality Reduction
Selects original variables	Creates new variables
Original meaning retained	New components may be harder to interpret
Example: Forward Selection	Example: PCA
Example: Backward Elimination	Example: LDA
Does not transform selected variables	Transforms original variables

---

54. PRINCIPAL COMPONENT ANALYSIS (PCA)

PCA is one of the most important dimensionality-reduction techniques.

Definition

PCA is an unsupervised linear transformation that converts correlated variables into a smaller number of uncorrelated principal components while preserving as much variance as possible.

---

55. PCA INTUITION

Suppose we have two highly correlated variables:

- Height
- Weight

The data may form an elongated cloud.

Instead of representing the data using both original variables, PCA finds a new direction that captures much of the variation.

The new variables are called:

Principal Components

---

56. PCA PROCEDURE

Step 1: Create Data Matrix

Represent observations as rows and features as columns.

Step 2: Center the Data

Subtract the mean of each feature.

Step 3: Scale if Appropriate

If variables are measured in very different units, standardization may be appropriate.

Step 4: Calculate Covariance Matrix

S=1n−1XTXS=\frac{1}{n-1}X^TX

for centered X.

Step 5: Calculate Eigenvalues and Eigenvectors

Eigenvectors give the principal directions.

Eigenvalues indicate the amount of variance associated with those directions.

Step 6: Sort Components

Sort eigenvalues from largest to smallest.

Step 7: Select Components

Choose the first k components.

Step 8: Transform Data

Project original observations onto these components.

---

57. PCA EXPLAINED VARIANCE

Suppose eigenvalues are:

Component	Eigenvalue

PC1	8
PC2	5
PC3	2
PC4	1

Total:

8+5+2+1=168+5+2+1=16

Explained variance of PC1:

816=50%\frac{8}{16}=50\%

PC1 + PC2:

8+516=81.25%\frac{8+5}{16}=81.25\%

Therefore, the first two components explain **81.25%** of the total variance.

---

58. ADVANTAGES OF PCA

1. Reduces dimensionality.
2. Removes linear correlation between components.
3. Can reduce computational cost.
4. Useful for visualization.
5. Can reduce redundancy.
6. Can provide compact representations.

---

59. LIMITATIONS OF PCA

1. Components can be difficult to interpret.
2. PCA is unsupervised.
3. PCA maximizes variance, not necessarily predictive performance.
4. Scaling can significantly affect results.
5. Sensitive to unusual observations.
6. It primarily captures linear structure.

---

60. LINEAR DISCRIMINANT ANALYSIS (LDA)

LDA is a **supervised** dimensionality-reduction method.

Unlike PCA, LDA uses class labels.

Objective

LDA attempts to find projections that:

- Maximize separation between classes.
- Minimize variation within each class.

---

61. PCA VS LDA

PCA	LDA
Unsupervised	Supervised
Does not use class labels	Uses class labels
Maximizes variance	Maximizes class separation
Useful for compression	Useful for classification-oriented projection
Components based on variance	Components based on class discrimination

Example

Suppose there are three classes:

- Class A
- Class B
- Class C

LDA can produce at most:

$C-1=3-1=2$C-1=3-1=2

useful discriminant dimensions.

---

62. LDA PROCEDURE

1. Calculate class means.
2. Calculate overall mean.
3. Calculate within-class scatter matrix.
4. Calculate between-class scatter matrix.
5. Find discriminant directions.
6. Select useful directions.
7. Project data onto those directions.

---

63. SINGULAR VALUE DECOMPOSITION (SVD)

SVD is an important matrix decomposition technique.

For a matrix A:

A=UΣVTA=U\Sigma V^T

where:

- **U** = left singular vectors
- **Σ** = singular values
- **V^T** = transpose of right singular vectors

---

64. SVD FOR DIMENSIONALITY REDUCTION

Suppose:

A=UΣVTA=U\Sigma V^T

The singular values in Σ are arranged from largest to smallest.

If only the largest k singular values are retained:

Ak=UkΣkVkTA_k=U_k\Sigma_kV_k^T

This gives a lower-rank approximation.

Example

Suppose a matrix has 100 dimensions.

Instead of retaining all 100 singular values, we may retain the first 10.

This produces a compact representation.

---

65. APPLICATIONS OF SVD

SVD is used in:

- Dimensionality reduction
- Image compression
- Noise reduction
- Recommendation systems
- Text analysis
- Latent Semantic Analysis
- Matrix approximation
- Numerical linear algebra

---

66. RELATIONSHIP BETWEEN PCA AND SVD

PCA can be computed using SVD.

For a centered data matrix:

X=UΣVTX=U\Sigma V^T

The columns of V represent principal directions.

The singular values determine the amount of variance captured by those directions.

Therefore:

SVD = General Matrix Decomposition

PCA = Statistical Dimensionality-Reduction Method

SVD can be used as an efficient computational method for PCA.

67. COMPARISON: FORWARD, BACKWARD, PCA, LDA AND SVD

Technique	Main Purpose	Uses Labels?	Output
Forward Selection	Feature selection	Usually model-dependent	Original selected features
Backward Elimination	Feature selection	Usually model-dependent	Original selected features
PCA	Dimensionality reduction	No	Principal components
LDA	Supervised dimensionality reduction	Yes	Discriminant components
SVD	Matrix factorization	No inherent labels	U, Σ, V^T

68. IMPORTANT FORMULAS – QUICK REFERENCE

Mean

x¯=∑xin\bar{x}=\frac{\sum x_i}{n}

Variance

s2=∑(xi−x¯)2n−1s^2=\frac{\sum(x_i-\bar{x})^2}{n-1}

Standard Deviation

s=s2s=\sqrt{s^2}

IQR

IQR=Q3−Q1IQR=Q_3-Q_1

Standardization

z=x−μσz=\frac{x-\mu}{\sigma}

Min-Max Scaling

x′=x−xminxmax−xminx'=\frac{x-x_{min}}{x_{max}-x_{min}}

Covariance

Cov(X,Y)=∑(xi−x¯)(yi−y¯)n−1Cov(X,Y)=\frac{\sum(x_i-\bar{x})(y_i-\bar{y})}{n-1}

Correlation

r=Cov(X,Y)sXsYr=\frac{Cov(X,Y)}{s_Xs_Y}

Bayes' Theorem

P(A∣B)=P(B∣A)P(A)P(B)P(A|B)=\frac{P(B|A)P(A)}{P(B)}

Linear System

Ax=bAx=b

PCA Covariance Matrix

S=1n−1XTXS=\frac{1}{n-1}X^TX

SVD

A=UΣVTA=U\Sigma V^T

---

69. COMPLETE CONCEPTUAL FLOW

The complete relationship between the topics can be understood as:

Raw Data

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Data Collection

↓

Data Cleaning

↓

Data Preprocessing

↓

Descriptive Statistics

↓

Univariate Analysis

↓

Bivariate Analysis

↓

Multivariate Analysis

↓

Feature Engineering

↓

Feature Selection

↓

Dimensionality Reduction

↓

PCA / LDA / SVD

↓

Machine Learning Model

↓

Evaluation

↓

Deployment

Quick Memory Points

Machine Learning: Learn patterns from data.

Supervised: Labeled data.

Unsupervised: Unlabeled data.

Big Data: Volume + Velocity + Variety + Veracity + Value.

Descriptive: What happened?

Diagnostic: Why did it happen?

Predictive: What is likely to happen?

Prescriptive: What action should be considered?

Univariate: One variable.

Bivariate: Two variables.

Multivariate: Multiple variables.

Feature Selection: Select original features.

PCA: Maximum variance.

LDA: Maximum class separation.

SVD: Matrix factorization.

Gaussian Elimination: Solve linear systems.

Probability: Quantifies uncertainty.

Statistics: Understands and summarizes data and supports inference.